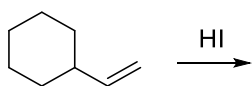

The Addition of HCl to Carvone

Name: _____

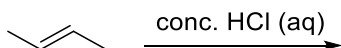
1) What is the name of the **reverse reaction** of an HX (e.g. HBr) addition to an alkene? (1 pt)

2) Is the addition of H-Br to an alkene entropically favored or disfavored? Explain why in 1-2 sentences. (2 pts)

3) What is the **major product** formed from the addition of HI to vinylcyclohexane (shown below)? (2 pts)

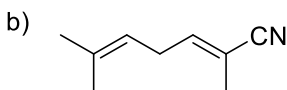
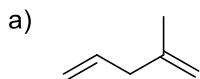


4) A chemist attempts to chlorinate *E*-2-butene using concentrated, aqueous HCl solution fails to isolate any 2-chlorobutane. What undesired product would be formed instead? (2 pts)



5) The organic compounds below each contain two alkenes. In each example:

- Circle the individual alkene that would **react fastest** with an equivalent of HCl. (2 pts)
- Within the alkene that reacts the fastest, use an asterisk (*) to indicate which carbon chlorine would add to. (2 pts)



- 6) Alkenes react with hypobromous acid (HOBr) to generate halohydrins and proceed through a very similar mechanism as HX additions (Hint: Br is the electrophile in this reaction). Although two regioisomers are possible, only one is formed. Select the **major isomer** and **rationalize** your choice. (4 pts)

